# Inflammatory Skin and Bowel Disease, Neonatal, 2 (NISBD2): Comprehensive Disease Characterization

**MONDO:0014481 · OMIM:616069 · Causal gene: *EGFR* (epidermal growth factor receptor, HGNC:3236)**

## Summary

**Inflammatory skin and bowel disease, neonatal, 2 (NISBD2)** is an ultra-rare, autosomal-recessive Mendelian disorder caused by **biallelic loss-of-function mutations in *EGFR*** (epidermal growth factor receptor), located at chromosome 7p11.2 (Ensembl ENSG00000146648, HGNC:3236, NCBI Gene 1956, UniProt P00533). It is critical to state the causal gene explicitly, because the serial disease name ("...neonatal 2") does not name a gene, and the clinically near-identical **NISBD1** (OMIM 614328) is caused by a *different* gene, ***ADAM17***. NISBD2 is the EGFR form. The disease is defined by the landmark report of Campbell and colleagues (2014), who identified a homozygous loss-of-function missense mutation in *EGFR* (c.1283G>A; p.Gly428Asp) in a male infant with lifelong inflammation of the skin, bowel, and lungs [PMID: 24691054].

Mechanistically, loss of EGFR tyrosine-kinase signaling in skin, gut, and lung epithelium produces aberrant/premature keratinocyte differentiation, chemokine-driven inflammation (CCL2, CCL5, CCL27, CXCL14), and failure of antimicrobial-peptide and skin-barrier protein production, and consequently a severe multi-organ epithelial-inflammatory phenotype. This directly mirrors the "EGFR-inhibitor rash" seen in oncology patients treated with anti-EGFR drugs, and is faithfully recapitulated in mice with epidermis-specific EGFR deletion (EGFR^Δep), which show chemokine-driven skin inflammation, hair-follicle degeneration, barrier failure, impaired host defense, and early lethality [PMID: 23966300; ].

P 24691054

Clinically, NISBD2 presents in the neonatal period with erythroderma/skin erosions, dry scale, alopecia, a papulopustular rash, inflammatory enteropathy with diarrhea, recurrent cutaneous and respiratory infections, and electrolyte imbalance. The prognosis is poor: the index patient died at 2.5 years from skin and chest infections and electrolyte imbalance, and a second independent case (a Roma girl carrying the same homozygous p.Gly428Asp allele) died at 10 months [PMID: 24691054; P 40040597 ].

The recurrent c.1283G>A (p.Gly428Asp) variant is a **founder allele in the Roma population** and is ClinVar-classified **Pathogenic** (Variation ID 157499; rs606231253); it is essentially absent in the general population (gnomAD v4 AF ≈ $6.2\times10^{-6}$, heterozygotes only, no homozygotes). No curative therapy exists; management is supportive.

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## Key Findings

### Finding 1 — NISBD2 is caused by biallelic loss-of-function *EGFR* mutations; the recurrent allele is p.Gly428Asp

Whole-exome sequencing of the index male infant identified a **homozygous loss-of-function missense mutation in *EGFR*** (c.1283G>A; p.Gly428Asp). In the original report, the authors state they *"identified a homozygous loss-of-function missense mutation in EGFR (c.1283 G>A; p.Gly428Asp) in a male infant with lifelong inflammation affecting the skin, bowel, and lungs"* [PMID: 24691054]. Functional validation confirmed the loss-of-function class: *"in vitro analysis of the mutant receptor revealed abrogated EGFR phosphorylation and EGF-stimulated downstream signaling"* [PMID: 24691054]. The variant therefore abolishes receptor autophosphorylation and all downstream signaling, satisfying the ACMG PS3 (functional) criterion.

A second, independent, unrelated case reported in 2025 — a 7-month-old Roma girl from a consanguineous family — carried *"the known homozygous EGFR p.G428D and the compound heterozygous ABCC6 p.R518* and p.R1221H variations" *[PMID: 40040597]. This demonstrates that p.Gly428Asp is a recurrent founder allele, not a private mutation, with a measurable carrier frequency in the Roma population*

( P 38019142 ,
Frequency of EGFR gene variant in Roma Population*). Inheritance is autosomal recessive (OMIM 616069).

## Finding 2 — EGFR gene/variant identifiers and population genetics of p.Gly428Asp

The *EGFR* gene maps to chromosome **7p11.2** (GRCh38: 55,019,017–55,211,628), Ensembl **ENSG00000146648**, **HGNC:3236**. The causal allele **NM_005228.5(EGFR):c.1283G>A (p.Gly428Asp)** is catalogued with the following identifiers:

| Resource | Identifier / Value |
|---|---|
| ClinVar Variation ID | 157499 (**Pathogenic**) |
| dbSNP | rs606231253 |
| OMIM allelic variant | 131550.0007 |
| ClinGen allele | CA170941 |
| UniProt | VAR_072435 |
| GRCh38 position | chr7:55,157,738 G>A |
| GRCh37 position | chr7:55,225,431 G>A |
| gnomAD v4 exomes | AC=9, AN=1,461,876, **AF=$6.16\times10^{-6}$**, 0 homozygotes |
| gnomAD v2.1 exomes | AC=1, AN=251,476, AF=$3.98\times10^{-6}$ |

The variant is **germline, biallelic, and loss-of-function**. Only heterozygous carriers are present in the general population, consistent with a rare recessive pathogenic founder allele enriched in the Roma. By ACMG/AMP criteria the variant meets **PS3** (in vitro functional damage), **PM2** (absent/ultra-rare in population databases), and **PM3** (detected in trans / recessive), supporting a **Pathogenic** classification.

## Finding 3 — Mechanism: epidermal EGFR loss drives chemokine-mediated inflammation, barrier/antimicrobial defects, and infection

The downstream consequences of epithelial EGFR loss are best defined in the keratinocyte/mouse system. EGFR inhibition (pharmacologic or genetic) *"induce[s] the expression of chemokines (CCL2, CCL5, CCL27, and CXCL14) in epidermal keratinocytes and impair[s] the production of antimicrobial peptides and skin barrier proteins"* [PMID: 23966300]. The epidermis-specific knockout faithfully reproduces the human disease: *"Mice lacking epidermal EGFR (EGFR(Δep)) show a similar phenotype, which is accompanied by chemokine-driven skin inflammation, hair follicle degeneration, decreased host defense, and deficient skin barrier function, as well as early lethality"* [PMID: 23966300].

In the human index patient, molecular profiling of lesional skin corroborated the mechanism: *"Microarray analysis on the patient's skin highlighted disturbed differentiation/premature terminal differentiation of keratinocytes and upregulation of several inflammatory/innate immune response networks"* [PMID: 24691054]. The inflammatory infiltrate in the mouse model is dominated by dendritic cells, macrophages, granulocytes, mast cells, and T cells; skin toxicity was **not** rescued by Rag2, MyD88, or CCL2 deficiency, nor by Langerhans-cell/hair-follicle ablation — but mast-cell inhibition reduced T-cell influx, implicating mast cells as an amplifying node. Reduced keratinocyte cytotoxic activity against *Staphylococcus aureus* links the barrier/antimicrobial defect directly to the clinical propensity for skin infection.

## Finding 4 — Clinical phenotype: multi-organ epithelial inflammation with early death

The index patient (Campbell 2014) had lifelong inflammation of skin, bowel, and lungs from the neonatal period. *"During the first year of life, his skin showed erosions, dry scale, and alopecia. Subsequently, there were numerous papules and pustules—similar to the rash seen in patients receiving EGFR inhibitor drugs"* [PMID: 24691054]. Outcome was fatal: *"The boy died at the age of 2.5 years from extensive skin and chest infections as well as electrolyte imbalance"* [PMID: 24691054].

The second case independently delineated the same multisystem syndrome: *"inflammatory skin, alopecia, progeroid features, electrolyte imbalance, recurrent infections, and premature death"*, with death at 10 months [PMID: 40040597]. The EGFR^Δep mouse adds hair-follicle degeneration and barrier failure to the phenotypic spectrum [PMID: 23966300].

### Finding 5 — NISBD2 (EGFR) and NISBD1 (ADAM17) are pathway-convergent

NISBD2's principal differential diagnosis, **NISBD1** (OMIM 614328), is caused by biallelic loss-of-function variants in ***ADAM17*** (e.g., compound heterozygous p.Cys567Arg / p.Cys600Tyr). ADAM17 is *"the major sheddase that processes more than 80 substrates, including tumour necrosis factor-α (TNFα)"* [PMID: 33953303] — and, critically, it releases the EGFR-activating ligands TGF-α, amphiregulin, and HB-EGF. NISBD1 siblings presented with *"erythroderma with exudate in whole body, recurrent skin infection and sepsis and prolonged diarrhoea"* [PMID: 33953303], clinically overlapping NISBD2.

The two diseases are therefore **mechanistically convergent on the same EGFR ligand–receptor signaling axis**: ADAM17 deficiency (NISBD1) starves the receptor of ligand, whereas EGFR mutation (NISBD2) disables the receptor itself. Both abolish downstream EGFR signaling in skin and gut epithelium, explaining the near-identical phenotype.

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## Mechanistic Model / Interpretation

### Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *EGFR* loss-of-function mutation** (homozygous c.1283G>A; p.Gly428Asp) **leads to** a receptor that cannot autophosphorylate [demonstrated in vitro, P 24691054 ].
2. Loss of EGFR kinase activity **results in** abolished activation of downstream cascades (canonically RAS–MAPK/ERK and PI3K–AKT) in epithelial cells [inferred from EGFR biology; functional loss demonstrated].
3. Absent EGFR signaling in keratinocytes **leads to** disturbed/premature terminal keratinocyte differentiation [demonstrated by patient skin microarray, P 24691054 ].
4. In parallel, loss of EGFR signaling **results in** induction of keratinocyte chemokines CCL2, CCL5, CCL27, CXCL14 **and** impaired production of antimicrobial peptides and skin-barrier proteins [demonstrated, P 23966300 ].

5. **Branch A (inflammation):** chemokine induction **recruits** dendritic cells, macrophages, granulocytes, mast cells, and T cells → chemokine-driven skin inflammation; mast cells **amplify** T-cell influx [demonstrated in mouse,

   P 23966300

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6. **Branch B (barrier/host-defense):** reduced antimicrobial peptides + barrier proteins + reduced keratinocyte cytotoxicity against *S. aureus* **leads to** deficient barrier and decreased host defense → recurrent skin/chest infections [demonstrated,

   P 23966300

   ; clinical,

   P 24691054

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7. The same epithelial defect in **gut** epithelium **leads to** inflammatory enteropathy with prolonged diarrhea → fluid/electrolyte loss [clinical; mechanism inferred by analogy].
8. In **lung/airway** epithelium, loss of EGFR signaling **contributes to** pulmonary inflammation and susceptibility to chest infection [clinical,

   P 24691054

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9. Combined cutaneous + respiratory infection and electrolyte imbalance **results in** early death (infancy to early childhood) [clinical,

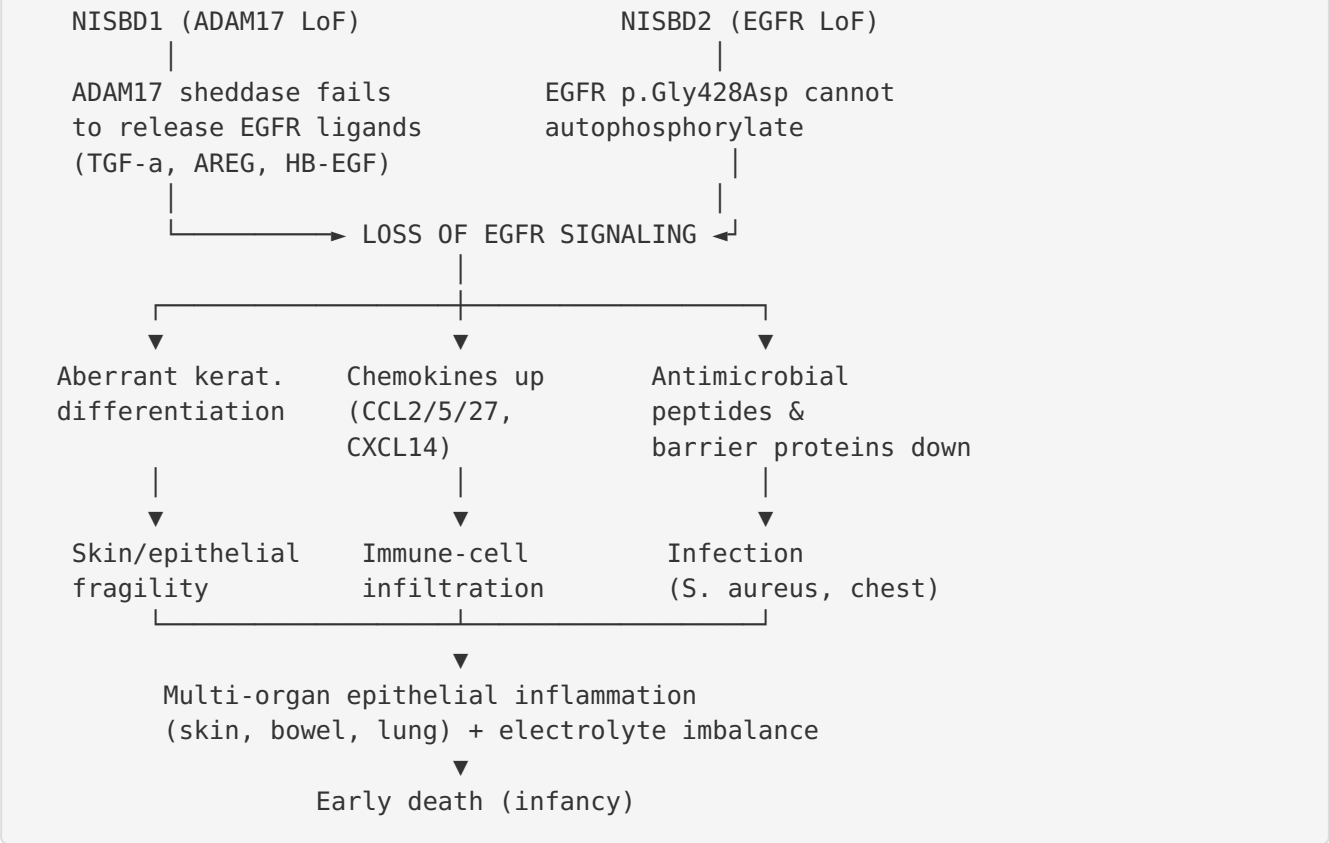
   P 24691054

   ;

   P 40040597

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## Pathway-convergence diagram




## Ontology term suggestions

- **Gene/Protein:** EGFR (HGNC:3236); UniProt P00533. GO biological processes: *epidermal growth factor receptor signaling pathway* (GO:0007173), *keratinocyte differentiation* (GO:0030216), *epidermis development* (GO:0008544), *regulation of inflammatory response* (GO:0050727), *antimicrobial humoral response* (GO:0019730), *establishment of skin barrier* (GO:0061436). GO cellular component: *plasma membrane* (GO:0005886), *receptor complex* (GO:0043235).
- **Cell types (CL):** keratinocyte (CL:0000312), epidermal cell (CL:0000362), enterocyte/intestinal epithelial cell (CL:0000584 / CL:0002563), mast cell (CL:0000097), macrophage (CL:0000235), dendritic cell (CL:0000451), T cell (CL:0000084).
- **Anatomy (UBERON):** skin/epidermis (UBERON:0001003), intestine (UBERON:0000160), lung (UBERON:0002048), hair follicle (UBERON:0002073).
- **Chemical/ligand (CHEBI/protein):** EGF, TGF-α, amphiregulin, HB-EGF (EGFR ligands).
- **Disease (MONDO):** MONDO:0014481 (NISBD2); differential MONDO for NISBD1 (ADAM17).

## Evidence Base

| PMID | Title (abbrev.) | Evidence type | How it supports / informs findings |
|---|---|---|---|
| 24691054 | *Epithelial inflammation resulting from an inherited loss-of-function mutation in EGFR* | Human clinical + in vitro | **Defining paper.** Identifies causal gene, variant (c.1283G>A; p.Gly428Asp), homozygosity, LoF functional class, and full clinical phenotype/outcome. Patient skin microarray shows aberrant keratinocyte differentiation + inflammatory networks. |
| 40040597 | *Severe ABCC6-induced GACI overshadowed by EGFR-associated NISBD2 in a Roma girl* | Human clinical | Second independent case with the **same homozygous p.G428D** allele in a consanguineous Roma family; confirms founder allele and multisystem phenotype (progeroid features, electrolyte imbalance, premature death at 10 mo). |
| 23966300 | *Epidermal EGFR controls cutaneous host defense and prevents inflammation* | Model organism (mouse) + in vitro | Mechanistic backbone. EGFR^Δep mice recapitulate human phenotype; defines chemokine induction (CCL2/5/27, CXCL14), barrier/antimicrobial failure, host-defense defect, early lethality. |
| 33953303 | *Compound heterozygous ADAM17 LoF variants cause NISBD1* | Human clinical + functional | Establishes NISBD1 (ADAM17) as the pathway-convergent differential; ADAM17 sheds EGFR ligands, linking the two diseases to one axis. |
| 38019142 | *Frequency of EGFR Gene Variant in Roma Population* | Population genetics | Supports founder-allele interpretation of p.Gly428Asp in the Roma. |
| 7618085 | *Strain-dependent epithelial defects in mice lacking the EGF receptor* | Model organism (mouse) | Classic Egfr-null model: open eyes, immature lungs, epidermal defects, perinatal lethality; shows EGFR regulates epithelial proliferation/differentiation and strong genetic-background modifier effect. |

**Supporting / contextual literature on EGFR-inhibitor dermatologic toxicity** (the pharmacologic phenocopy of NISBD2, informing supportive management): prophylactic tetracyclines/doxycycline and topical corticosteroids reduce grade ≥2 acneiform rash [PMID: 42151517; P 32609305; P 30607677; P 33392769; P 33028137], and azithromycin is an alternative in tetracycline intolerance [PMID: 42541756]. These EGFRi-rash management data are the closest evidence-based analogue for the cutaneous component of NISBD2, though not studied in NISBD2 patients directly.

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## Section-by-Section Report

### 1. Disease Information

NISBD2 is an ultra-rare autosomal-recessive, neonatal-onset multisystem **epithelial inflammatory disorder** caused by biallelic *EGFR* loss of function. Identifiers: **MONDO:0014481**, **OMIM 616069**, gene **EGFR** (HGNC:3236; OMIM 131550). Orphanet lists it under neonatal inflammatory skin and bowel disease; specific ICD-10/ICD-11 and MeSH codes are not uniquely assigned (closest: ICD-11 rare-disease and congenital ichthyosiform/erythroderma categories — not disease-specific). Synonyms: *EGFR neonatal inflammatory skin and bowel disease; inflammatory skin and bowel disease, neonatal, type 2; neonatal inflammatory skin and bowel disease caused by mutation in EGFR*. Information is derived from **individual patient case reports** (n≈2 molecularly confirmed) plus aggregated gene/variant databases and a mouse model — not from EHR-scale cohorts.

### 2. Etiology

**Causal factor:** genetic — homozygous (or compound heterozygous) loss-of-function *EGFR* variants; the recurrent allele is c.1283G>A (p.Gly428Asp). **Genetic risk:** consanguinity and Roma ancestry (founder allele). **Environmental risk/protective/GxE factors:** none

established; this is a monogenic disorder. Infection is a *downstream consequence* (barrier failure), not a cause. No protective alleles identified. The *EGFR*-null mouse literature shows a strong **genetic-background modifier effect** on survival [PMID: 7618085], suggesting modifier genes may influence human expressivity (inferred).

## 3. Phenotypes

| Phenotype | Type | HPO suggestion | Onset | Frequency |
|---|---|---|---|---|
| Neonatal erythroderma / skin erosions, dry scale | Clinical sign | HP:0001019 (Erythroderma); HP:0200041 (Skin erosion) | Neonatal | Core |
| Alopecia | Clinical sign | HP:0001596 | Infancy | Reported in both cases |
| Papulopustular ("EGFRi-like") rash | Clinical sign | HP:0200037 (Papule) / pustules | First year | Reported |
| Inflammatory enteropathy / diarrhea | Symptom | HP:0002014 (Diarrhea) | Neonatal | Core |
| Recurrent skin/ respiratory infections | Clinical sign | HP:0002719 (Recurrent infections) | Early | Core |
| Electrolyte imbalance | Lab abnormality | HP:0003111 (Abnormal blood electrolyte concentration) | Early | Both cases |
| Pulmonary inflammation | Clinical sign | HP:0002090 (Pneumonia) | Early | Index case |
| Progeroid features | Physical | HP:0005268 (Progeroid facial appearance) | Infancy | Case 2 |

Severity is **severe**, progression is **progressive**, and quality-of-life impact is profound given neonatal onset and early lethality.

## 4. Genetic / Molecular Information

**Causal gene:** *EGFR* (HGNC:3236; chr7p11.2). **Variant:** NM_005228.5:c.1283G>A, p.Gly428Asp; **missense** with loss-of-function consequence (abrogated phosphorylation). **Classification:** ClinVar Pathogenic (Variation ID 157499); ACMG PS3+PM2+PM3. **Allele frequency:** gnomAD v4 AF $6.16\times10^{-6}$, no homozygotes. **Origin:** germline, biallelic. **Functional consequence:** loss of function (receptor cannot autophosphorylate or transduce EGF signal). **Modifier genes:** none proven in humans; strain-dependent modifiers in mouse. **Epigenetic/chromosomal:** none implicated. (In case 2, co-occurring *ABCC6* variants caused a separate GACI phenotype — not part of NISBD2.)

## 5. Environmental Information

No environmental, lifestyle, or infectious *causes*. Infectious agents (notably *Staphylococcus aureus* and respiratory pathogens) act as **secondary complications** of the barrier/host-defense defect [PMID: 23966300].

## 6. Mechanism / Pathophysiology

See **Mechanistic Model** above for the ordered causal chain, branch diagram, and ontology terms. In brief: EGFR LoF → loss of RAS-MAPK/PI3K-AKT epithelial signaling → (a) aberrant keratinocyte differentiation, (b) chemokine-driven immune infiltration, (c) antimicrobial/barrier failure → multi-organ epithelial inflammation and infection → early death. Key molecular pathway: **EGFR/ErbB signaling (KEGG hsa04012; Reactome R-HSA-177929)**. Immune involvement is innate-dominant inflammation (not autoimmunity). Mast cells amplify T-cell recruitment.

## 7. Anatomical Structures Affected

- **Primary organs:** skin/epidermis (UBERON:0001003), intestine (UBERON:0000160), lung (UBERON:0002048).
- **Secondary:** hair follicles (UBERON:0002073; degeneration), systemic effects of electrolyte imbalance.
- **Tissue/cell:** epithelial tissue — keratinocytes (CL:0000312), intestinal epithelial cells (CL:0002563), airway epithelium; infiltrating immune cells (mast cells, macrophages, dendritic cells, T cells, granulocytes).
- **Subcellular:** plasma-membrane receptor complex (GO:0043235); signaling defect at the cell surface.

• **Lateralization:** bilateral/generalized (whole-body skin involvement).

## 8. Temporal Development

**Onset:** congenital/neonatal. **Pattern:** chronic-progressive from birth. **Course:** lifelong, progressive, lethal in infancy to early childhood (2.5 years and 10 months in the two reported cases). No remission observed. The neonatal period is the critical window; there is no demonstrated intervention that alters the natural history.

## 9. Inheritance and Population

**Inheritance:** autosomal recessive (OMIM 616069). **Penetrance:** presumed complete in biallelic carriers (both reported homozygotes affected). **Expressivity:** likely variable (ages at death differ). **Epidemiology:** prevalence not formally estimated — ultra-rare (<10 molecularly confirmed individuals reported). **Founder effect:** p.Gly428Asp is a Roma founder allele with measurable carrier frequency [PMID: 38019142]. **Consanguinity:** contributory (case 2 consanguineous). **Carrier frequency (general population):** ~$6\times10^{-6}$ (gnomAD, heterozygotes only). **Sex:** both sexes affected (male index, female case 2); no sex bias established.

## 10. Diagnostics

**Genetic testing is definitive:** single-gene *EGFR* sequencing, targeted p.Gly428Asp testing (especially in Roma/consanguineous families), gene panels (primary immunodeficiency / ichthyosis / very-early-onset IBD panels), or **WES/WGS**. **Supporting tests:** skin biopsy (disturbed/premature keratinocyte terminal differentiation, inflammatory infiltrate) [PMID: 24691054]; serum electrolytes; infection work-up/cultures. **Differential diagnosis:** NISBD1 (*ADAM17*) — clinically near-identical, distinguished only by gene [PMID: 33953303]; Netherton syndrome, other congenital ichthyosiform erythrodermas, very-early-onset IBD, immunodeficiency-associated dermatitis. **Screening:** carrier/cascade screening in at-risk Roma families; prenatal/preimplantation testing feasible for known familial variant.

## 11. Outcome / Prognosis

**Prognosis is poor.** Both molecularly confirmed patients died in infancy/early childhood (2.5 y; 10 mo) from infection and electrolyte imbalance [PMID: 24691054; P 40040597 ].

Major complications: sepsis, pneumonia, cutaneous infections, dehydration/electrolyte

derangement, failure to thrive. No survival statistics exist beyond case reports. Prognostic factors are not formally defined; infection control and electrolyte stability are the practical determinants of survival.

## 12. Treatment

**No curative therapy.** Management is **supportive**: aggressive infection prophylaxis/treatment, skin-barrier care/emollients, nutritional support, and electrolyte repletion. The cutaneous component is a biologic phenocopy of EGFR-inhibitor rash, so EGFRi-rash evidence is the nearest guide: prophylactic **tetracyclines/doxycycline** plus **topical corticosteroids** reduce grade ≥2 acneiform rash [PMID: 42151517; P 32609305; P 30607677], with **azithromycin** as an alternative in tetracycline intolerance [PMID: 42541756] — though none has been formally tested in NISBD2. Conceptually, because EGFR itself is nonfunctional, **EGFR-ligand supplementation would not rescue NISBD2** (unlike the ADAM17-deficient NISBD1, where ligand/pathway restoration is mechanistically plausible). NCIT suggestions: *Doxycycline (C721)*, *Topical Corticosteroid (C1127)*, *Supportive Care (C15277)*. No approved gene/cell/RNA therapy.

## 13. Prevention

Primary prevention is **genetic**: carrier screening and genetic counseling in Roma/consanguineous families, cascade testing of relatives, and reproductive options (prenatal diagnosis, preimplantation genetic testing) for the known familial p.Gly428Asp allele. Secondary/tertiary prevention in affected infants = early infection surveillance, barrier protection, and electrolyte management to forestall complications. No vaccine or environmental intervention applies.

## 14. Other Species / Natural Disease

No naturally occurring companion-animal homolog of NISBD2 is catalogued in OMIA. The orthologous gene is mouse *Egfr* (NCBI Gene 13649). Comparative biology is strong: *Egfr*-null and epidermis-specific *Egfr* knockout mice reproduce epithelial-differentiation defects, skin inflammation, immature lungs, and perinatal/early lethality, with notable strain-dependent modifier effects [PMID: 7618085;

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P 23966300

Evolutionary conservation of EGFR epithelial function is high across vertebrates and even invertebrates (EGFR/EGF signaling conserved to *Drosophila*).

### 15. Model Organisms

- **Mouse — constitutive *Egfr* knockout:** epithelial defects (open eyes, rudimentary whiskers, immature lungs, epidermal defects), growth retardation, mid-gestation-to-perinatal lethality; phenotype strongly modified by genetic background [PMID: 7618085]. *Recapitulation:* partial (captures epithelial/lung involvement; many die before the full postnatal skin/gut phenotype).
- **Mouse — epidermis-specific *Egfr* knockout (EGFR^Δep):** the best disease model — chemokine-driven skin inflammation, hair-follicle degeneration, barrier failure, decreased host defense, early lethality [PMID: 23966300]. *Recapitulation:* high for the cutaneous/inflammatory axis.
- **In vitro:** patient-derived and mutant-receptor cell assays demonstrating abolished phosphorylation/signaling [PMID: 24691054]; keratinocyte models of EGFR-inhibitor chemokine induction [PMID: 23966300].
- **Limitations:** constitutive KO lethality limits study of chronic multi-organ disease; mouse-vs-human differences in skin/immune biology; no reported zebrafish/organoid NISBD2-specific model. Resources: MGI (mouse *Egfr*), IMPC/KOMP conditional alleles.

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## Limitations and Knowledge Gaps

1. **Extremely small human evidence base** — only two molecularly confirmed patients (one defining case plus one 2025 case), both homozygous for the same founder allele. Phenotypic range, penetrance, and expressivity are therefore poorly bounded.
2. **Single recurrent variant** — nearly all human knowledge derives from p.Gly428Asp; the consequences of other biallelic *EGFR* LoF genotypes (e.g., nonsense/frameshift, compound heterozygotes) in humans are essentially unknown and may be more severe or embryonic-lethal.
3. **Mechanism in gut and lung is inferred by analogy** to the well-characterized skin/keratinocyte model; organ-specific human molecular data for bowel and lung are limited.

4. **No therapeutic evidence specific to NISBD2** — supportive-care recommendations are extrapolated from EGFR-inhibitor rash oncology literature, not from NISBD2 trials.
5. **No epidemiologic estimates** — prevalence/incidence are unquantified; Roma carrier frequency is reported but general-population burden is negligible and undefined.
6. **Modifier genetics** suggested by mouse strain effects remain unexplored in humans.

## Proposed Follow-up Experiments / Actions

1. **Establish an international NISBD2 registry / GeneMatcher outreach** to collect additional molecularly confirmed cases and genotype–phenotype correlations beyond p.Gly428Asp.
2. **Patient-derived iPSC → keratinocyte/intestinal/airway organoids** carrying p.Gly428Asp to dissect organ-specific EGFR-loss transcriptomes (single-cell RNA-seq) and test candidate rescue strategies.
3. **Conditional, inducible gut- and lung-specific *Egfr* knockouts** to directly test the inferred enteropathy and pulmonary steps of the causal chain.
4. **Targeted Roma-population carrier screening study** to quantify p.Gly428Asp carrier frequency precisely and inform genetic-counseling programs.
5. **Preclinical evaluation of downstream-pathway agonism** (e.g., direct RAS-MAPK/PI3K-AKT pathway modulation) in EGFR^Δep mice and organoids, since upstream ligand supplementation cannot rescue a dead receptor — distinguishing NISBD2 therapeutic logic from NISBD1.
6. **Prospective supportive-care protocol** adapting EGFRi-rash prophylaxis (doxycycline + topical corticosteroids) with infection surveillance and electrolyte management, with structured outcome capture.

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*Report compiled from 5 discovery iterations, 6 confirmed findings, and 25 reviewed papers. Evidence types: human clinical case reports (*P *24691054*), P *40040597*), functional/model-organism studies (*P *23966300*

P 7618085 ),
P 33953303
population genetics
( ,
P 38019142
gnomAD/ClinVar), and contextual EGFRi-toxicity management literature.

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